

RESEARCH ARTICLE

A Multimodal Deep Fusion Method for Stock Movement Prediction Using Heterogeneous Data Source

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ABSTRACT Stock movement prediction remains challenging due to the complexity of integrating heterogeneous data sources and capturing intricate temporal-semantic relationships. Existing approaches suffer from incomplete semantic preservation, shallow fusion mechanisms, and inadequate multi-scale temporal modeling. In this paper, we propose a multimodal deep fusion framework that addresses these limitations through three synergistic components: 1) An SRTE module based on pretrained BERT that preserves 100% of textual content by segmenting long sequences and applying average pooling, recovering 17.82% more tokens than conventional truncation; 2) An MsLSTM architecture that simultaneously captures short-term price dynamics and long-term market trends through parallel processing of original and temporally smoothed sequences, and; 3) A CMCF mechanism that performs dual-level integration via across-temporal interval fusion and along-temporal step aggregation. We compare our proposed method against baseline models on the StockNet dataset, achieving superior predictive performance. Additionally, we conduct comprehensive ablation studies, back-testing analysis, long-text modeling comparisons, and interpretability analysis. The results validate the effectiveness of deep multimodal fusion for financial forecasting and provide actionable insights for data-driven decision-making in volatile market environments.

INDEX TERMS Financial time series forecasting, stock movement prediction, deep learning model, BERT, LSTM, heterogeneous feature embedding, multimodal fusion strategy.

I. INTRODUCTION

The stock market plays an important role in the world economy and has significant effects on personal economic life [1]. The ability to forecast the stock market is a difficult one for investors, financial analysts, and academic researchers [2], [3]. Accordingly, considerable efforts have been dedicated in academia to improve the ability to predict the stock market by using various theoretic perspectives,

statistical modeling methods, and machine learning techniques [4].

The majority of existing research has been based primarily on historical stock market data and derivative technical indicators (TI). Especially, deep learning techniques are widely accepted in the stock market prediction applications, ranging from individual models to complex ensemble-based structures. However, this methodology has some limitations, since it neglects the important underlying factors of the fundamental analysis [5]: political news, economic statistics, and sentiment of the market [4]. Therefore, the integration of a

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more diverse range of factors is necessary to improve the performance of stock market prediction [2].

The ubiquity of user-generated content on digital platforms yields large information assets that are of interest to businesses and firms for understanding consumer behavior, reputation management strategies, and identification of new markets to serve [6]. Recent developments in Natural Language Processing (NLP) technology have made it possible to integrate this various textual information into stock prediction models. Existing research widely adopts text embedding approaches to conduct text-to-numeric conversion on the textual contents (i.e., sentiment) that can be combined with the past price data. But it has a big drawback: it can hardly be applied to deep fusion for heterogeneous data. More specifically, prior methods are found to be insufficient to effectively learn about the extent to which latent factors hidden inside textual data drive historical market behaviors and, reciprocally, how historical market data affects investor sentiment that in turn leads to future user-generated content. Hence, there is an urgent demand for heterogeneous data fusion techniques that can improve stock market prediction.

In this study, we proposed an integrated predictive framework that utilized both market data and textual information. The proposed model can be divided into three main modules: a text data embedding module, a historical data embedding module, and a feature fusion module. The interrelation of these components enables the unified fusion of heterogeneous data. For the textual data embedding, we presented a method of the Sequence Reorganization Text Embedding method (SRTE) based on the pre-trained BERT Bidirectional Encoder Representation from Transformer (BERT) to retain the full length of textual embeddings with dimension alignment to the historical data representations. To capture multi-scale temporal patterns for historical data embedding, we employed a Multi-Scale LSTM (MsLSTM) architecture that processes the original sequence alongside a temporally smoothed version using 1D average pooling with stride = 1, enabling simultaneous extraction of short-term price dynamics and long-term market trends while maintaining strict time alignment and sequence length consistency. The feature fusion module proposes a novel Cross-Modal Cross-Fusion (CMCF) method, which can integrate the heterogeneous features by adopting a dual-level fusion mechanism: across-temporal interval fusion and along-temporal step aggregation. The method effectively addresses the ongoing issues of low data-utilization efficiency in heterogeneous data and insufficient fusion depth that existed in previous approaches.

To demonstrate the effectiveness of the proposed method, we employed multiple widely adopted classification task evaluation metrics, and we performed thorough ablation studies to illustrate the effectiveness of each atomic module for overall prediction performance. In addition, we compare the performance of our model with several baselines from the recent literature to show its competitive advantages.

The main contributions of this work are as follows:

- **Multi-modal information integration:** We utilize two groups of heterogeneous data sources (stock historical data and text data), where the stock historical data includes original five dimensions and multiple technical indices are added. In doing so, the information available to the model becomes enriched, and it is forced to learn over a broader set of valid features.
- **Late fusion strategy for holistic representation inclusion:** We adopted a late fusion approach for model design. To this end, we introduced a sequence rearrangement method to maintain the complete semantic information in the input text. For historical values, a multi-scale LSTM network was introduced, which learned more abundant context information so that the time series features can better predict high-frequency fluctuation in the short-term and the low-frequency trends in the long-term.
- **Novel deep fusion architecture:** We developed a novel CMCF method for the deep fusion of relevant feature representations. The proposed approach employs a hierarchical fusion strategy that involves a two-level process: first, it alternates the fusion of different feature vectors within individual temporal instances, and then it fuses these vectors across different temporal instances. This multi-stage structure overcomes the essential flaw caused by insufficient fusion depth.

The rest of the paper is organized as follows. Section II presents a literature review on related work. Section III introduces the materials and the proposed method in detail. Section IV introduces the development of the proposed method, including evaluation metrics, hyperparameter optimization, back-testing analysis, long-text modeling analysis, interpretability analysis, ablation study, and performance comparison. Section V provides discussions and future work directions.

II. LITERATURE REVIEW

Stock prediction constitutes a prominent research domain within time series analysis, which has received wide attention and research interest for its theoretical challenge and significant impact in practice. The existing literature reveals two types of methods that have emerged to address the challenges in stock prediction.

The first type of method uses traditional machine learning and deep learning frameworks that are based purely on the historical price data. These methods rely on sequential architectures that are particularly well suited to temporal dependencies, like recurrent neural networks (RNNs), LSTM networks, gated recurrent units (GRUs), and Transformer encoder architectures. Acknowledging the shortcomings of single-model architectures, subsequent studies have advanced to the construction of ensemble and hybrid models, which combine multiple models into a single system to address the limitations of single-model methods.

Another category of methods attempts to include high dimensionality of input features by using multi-modal data sources, in particular textual data, to enrich prediction models' predictions. This section synthesizes the current state of knowledge across both types of methods, reviewing integration strategies for historical data-based models and investigating the use of heterogeneous data fusion techniques in stock price forecasting applications.

The ensemble learning algorithm has been prevalent in machine learning when it comes to predicting stock price based on historical data. These approaches utilize machine learning algorithms to detect and extract patterns from stock market history [7]. Driven by the recent advances in deep learning, ensemble learning, which combines several models to analyze complex patterns in financial data, has become the most prevailing controller for stock market prediction [8]. The ensemble methods can take advantage of the complementarity between different algorithms so as to improve the predictive power for a particular task [9].

Liu et al. [10] introduced an attention-based bidirectional long short-term memory (Bi-LSTM) model to deal with non-smooth predictions and short-term dependencies, which are often amplified by the high noisiness of stock market data. In this context, the Bi-LSTM and attention mechanism play complementary roles through a coupled interaction. He et al. [11] proposed an ensemble method to effectively integrate linear and nonlinear features in predicting stock market data. This hybrid structure takes advantage of the capability of the autoregressive moving average (ARMA) model for capturing linear dependencies, while the convolutional neural network (CNN) and LSTM component models complex nonlinear patterns across temporal and spatial dimensions. Ho and Hung [12] presented a multivariate prediction method that integrates complete ensemble empirical mode decomposition and temporal fusion Transformer to cope with steady noise, non-linearities, non-smoothness, and ambiguities present in stock market data. Lazcano et al. [13] introduced an ensemble method integrating the advantages of graph CNNs and Bi-LSTM to do stock price prediction.

Heterogeneous data fusion methods represent a form of methods that consider how to integrate different source of information, typically combining quantitative historical price data and qualitative textual information in predictive models. These fusion approaches can be imaged based on their temporal integration architecture, leading to two main categories: early and late fusion techniques, characterized by their embedding procedure, and the multimodal integration level in the computational pipeline.

Early fusion of hybrid information refers to the integration of different modalities prior to their transformation into time series or semantic representations through embedding techniques [14]. This approach has become popular for contemporary financial time series prediction, particularly in stock forecasting [15], largely due to advancements in natural language processing. These developments have significantly improved the quantification of large volumes of unstructured

text data, which often contain complex and stochastic information, making them suitable for downstream predictive tasks. The early fusion method converts text data into sentiment scores as new variables that are compatible with historical data. For example, Deep [16] utilized TextBlob to do text embedding, while John and Latha [17] introduced a hybrid method combining the Valence Aware Dictionary and Sentiment Reasoner with a hybrid RNN to do the prediction. Liu et al. [18] proposed an ensemble architecture using Word to Vector for text embedding. However, these methods cannot capture contextual nuances and handle out-of-vocabulary words, which may limit their effectiveness in dynamic or domain-specific financial text. BERT [19], [29] and FinBERT [6] are also common methods for financial text embedding. Pre-trained BERT features a bidirectional modeling structure and task-specific fine-tuning, resulting in stronger contextual understanding compared to static word embedding approaches. FinBERT uses financial-specific text for pre-training. While the addition of sentiment scores as new variables can enhance prediction effectiveness, the heterogeneous information lacks deep interaction and integration.

Different from the early fusion method that converts text data into variables compatible with historical data, the late fusion method retains the complete semantic representation of the embedding, embeds it with the historical data, and then fuses it in the temporal dimension. Ronaghi et al. [20] proposed a method to fuse semantic representation utilizing Global Vectors for Word Representation and a bi-directional GRU to realize the deep interaction of multi-source information. Choi et al. [21] introduced a hybrid information mixing module that employs a late fusion strategy aimed at preserving the full semantic richness of pre-trained BERT's output. In this method, time-series features are extracted at each time step using GRU, while semantic features are obtained through BERT. These features are then fused and reused as inputs to the prediction model.

III. MATERIALS AND METHODS

A. DATA PREPROCESSING

The dataset employed in this study is StockNet [22], a comprehensive repository that integrates both historical stock data and textual data derived from social media platforms. The historical component of StockNet includes highly traded, independent stocks from nine distinct industries, sourced from the Standard & Poor's 500 index of the New York Stock Exchange and the NASDAQ market, as illustrated in Table 1. The textual data, collected under an official license from Twitter, is retrieved using regular expression queries constructed from NASDAQ ticker symbols, such as `"/$GOOG\b"` for Google Inc. [22].

When we preprocessed the data, we encountered missing data for three stocks (\$AGFS, \$BABA, and \$GMRE), making it impossible to generate a sample size corresponding to the text data. These three stocks were removed from the

TABLE 1. List of all 88 stock in stocknet dataset.

Industry	Symbols
Basic Materials	\$XOM, \$RDS-B, \$PTR, \$CVX, \$TOT, \$BP, \$BHP,
	\$SNP, \$SLB, \$BBL
Consumer Goods	\$AAPL, \$PG, \$BUD, \$KO, \$PM, \$TM, \$PEP, \$UN,
	\$UL, \$MO
Healthcare	\$JNJ, \$PFE, \$NVS, \$UNH, \$MRK, \$AMGN, \$MDT,
	\$ABBV, \$SNY, \$CELG
Services	\$AMZN, \$BABA, \$WMT, \$CMCSA, \$HD, \$DIS,
	\$MCD, \$CHTR, \$UPS, \$PCLN
Utilities	\$NEE, \$DUK, \$D, \$SO, \$NGG, \$AEP, \$PCG, \$EXC,
	\$SRE, \$PPL
Conglomerates	\$IEP, \$HRG, \$CODI, \$REX, \$SPLP, \$PICO, \$AGFS,
	\$GMRE
Financial	\$BCH, \$BSAC, \$BRK-A, \$JPM, \$WFC, \$BAC, \$V,
	\$C, \$HSBC, \$MA
Industrial Goods	\$GE, \$MMM, \$BA, \$HON, \$UTX, \$LMT, \$CAT,
	\$GD, \$DHR, \$ABB
Technology	\$GOOG, \$MSFT, \$FB, \$T, \$CHL, \$ORCL, \$TSM,
	\$VZ, \$INTC, \$CSCO

TABLE 2. Technical indicators used in this study.

Category	Indicator Name	References
Trend Indicators	MA 5/10/20	[23, 24, 26, 27, 28]
	MACD	[23, 24, 25, 26, 27, 28]
Oscillation Indicators	RSI	[24, 25, 26, 27, 28]
	William's %R	[23, 24, 25, 26, 27]
Momentum indicators	BB	[23, 26, 27]
	ROC	[23, 25, 27, 28]
	MOM	[23, 27, 28]
Volume Indicators	OBV	[23, 27]
	VPT	[27]

dataset we reconstructed by combining historical data and text data. There are a number of targets with exceptionally minor movement ratios. In a three-way stock trend prediction task, a common practice is to categorize these movements to another “preserve” class by setting upper and lower thresholds on the stock price change. Since we aim at the binary classification of stock changes identifiable from social media, Following the principle setting [22], we set two thresholds, -0.5% and 0.55%. Based on this threshold, different lag time (T^l) settings will result in different total number of samples and number of labels. When the lag time is 5 ($T^l = 5$), the total sample size is 42,415, of which 30.63% (12,991) are positive samples, 30.75% (13,042) are negative samples, 38.62% are neutral samples (16,382), and the ratio of the two sample labels is 0.996. When the lag time is 10 ($T^l = 10$), the total sample size is 41,990, of which 30.61% (12,854) are positive samples, 30.78% (12,926) are negative samples, 38.60% are neutral samples (16,210), and the ratio of the two sample labels is 0.994. With a lag time of 30 ($T^l = 30$), the total

sample size is 40290, of which 30.49% (12285) are positive samples, 30.69% (12365) are negative samples, 38.83% (15640) are neutral samples, and the ratio of the two sample labels is 0.994.

To achieve multi-scale information modeling and increase the feature expression ability of the model, we used comprehensive technical indicators that have appeared in previous studies. The financial technical indicators we use mainly include four categories: trend, oscillation, momentum, and volume. They are designed to allow the model to perceive the price structure at different time granularities and improve the modeling capabilities of short-, medium-, and long-term signals. Trend indicators include 5-day Moving Average (MA5), 10-day Moving Average (MA10), 20-day Moving Average (MA20), and Moving Average Convergence/Divergence (MACD). Oscillator indicators include Relative Strength Index (RSI), William's %R, and Bollinger Bands (BB). Momentum indicators include Rate of Change (ROC) and Momentum (MOM). Volume indicators include On-Balance Volume (OBV) and Volume Price Trend (VPT). Table 2 summarizes all the technical indicators used in our study and the literature on stock prediction in which they appear.

In the text preprocessing part, we processed the data returned by Twitter's raw API into clean natural language text. Specifically, we first retained only the main text part of each comment. In the second step, we cleaned the metadata contained in the original text part of the data, including retweet marks in the form of “RT@users”, URLs, extra characters, and leading and trailing spaces. In the third step, we retained emoticons, such as “\$” tags representing stock symbols and “#” tags representing topic tags, because these emoticons are useful for financial-related topic analysis.

Figure 1 presents a specific sample. This sample contains historical data for five variables spanning five trading days from January 2nd, 2014, to January 8th, 2014, for Apple Inc. Stock (\$AAPL), along with corresponding Twitter text data for each trading day, with a lag time of 5 ($T^l = 5$) is adopted. Twitter text data from non-trading days is incorporated into the next trading day's data. For example, in the given sample, Twitter text data from January 4th and 5th, 2014, are incorporated into January 6th, 2014. This is because the stock market does not trade during the holiday period, and the potential information in this text data will only be digested by the market and reflected in prices when the market opens on the next trading day.

The preprocessed dataset contains a total of 116,404 tweets spanning 85 stocks from January 2014 to January 2016. The tweet distribution varies across stocks, with an average of approximately 1,369 tweets per stock, reflecting different levels of market attention and social media activity. This substantial corpus of user-generated content, temporally aligned with daily historical price data, provides comprehensive textual information for multimodal fusion learning.

To ensure rigorous evaluation without lookahead bias, we adopt a time-respecting train-validation-test split that strictly preserves temporal ordering. The dataset is divided

historical component							textual component
Date	Open	High	Low	Close	Volume		
2014/1/2	79.382858	79.575714	78.860001	79.01857	58671200	→	\$ aapl - wall st . kicks off new year on lower note → stock stocks stockaction here's how apple could be making a huge push into healthcare ...
2014/1/3	78.980003	79.099998	77.204285	77.28286	98116900	→	our top 3 trade ideas for 2014 and two \$ 1,000 futures wins already in the bank \$ gld \$ uso \$ aapl -- the five-year-old ipad : when will people stop upgrading? ...
2014/1/6	76.778572	78.114288	76.228569	77.704285	103152700	→	in it to win it : \$ mu puts working .. \$ aapl long working .. \$ san sitting : in it to win it \$ aapl intel has a shot at winning apple wall street vs aapl in nine bar-charts ...
2014/1/7	77.760002	77.994286	76.845711	77.148575	79302300	→	\$ aapl i love my ipad b / c no virus and i am protected by apple . android is open source and full of virus etc . i don't want it in my car ...
2014/1/8	76.972855	77.937141	76.955711	77.637146	64632400	→	\$ aapl app store sales > \$ 10 billion digital iphone ipad mac revenue profit growth tech lob itunes usa new deal , new stores : apple' s plan for china via \$ aapl \$ aapl over 546.86 ...

FIGURE 1. A specific sample containing historical and textual component for Apple Inc. Stock (\$AAPL) from January 2, 2014, to January 8, 2014.

chronologically into training (first 70%), validation (next 15%), and testing (last 15%) sets. This ensures the model is trained exclusively on historical data and evaluated on strictly future unseen data. Explicit leakage controls are implemented throughout the pipeline: (1) our MsLSTM employs causal padding, where smoothed features at time t use only observations from $[t-k+1, t]$; (2) all technical indicators are computed sequentially using only past values; (3) no data shuffling or stratified sampling is performed that could violate temporal causality.

We formulate stock prediction as a binary classification task (movement prediction) rather than regression (price prediction) for several reasons. First, classification is more robust to the extreme volatility and non-stationarity inherent in financial markets. Second, textual sentiment from social media naturally aligns with directional signals rather than specific price targets. This formulation is also consistent with recent literature [21], [22] and enables direct comparison with established baselines.

B. PROPOSED METHOD

Fig. 2 illustrates the procedure of the proposed method, which is based on a two-pathway embedding strategy, where the text input and the history input are embedded separately before being combined. This design keeps the dimensionality of the textual embedding unchanged so that the vector representation capability after embedding is guaranteed. The methodology comprises three distinct modules, each of which is elaborated upon in subsequent sections.

1) **SEQUENCE REORGANIZATION TEXT EMBEDDING-BERT** [29] serves as a base embedding model for text data input due to the powerful dynamic embedding ability and high-dimensional semantic representation, and it can provide rich semantic information for heterogeneous feature fusion. But during the embedding process, there is a high limit of 512 tokens as the sequence length to the pre-trained BERT model. However, the preceding cutoff will be suboptimal when the preprocessed texts are long and surpass the token capacity of BERT (e.g., 512 tokens), where semantic

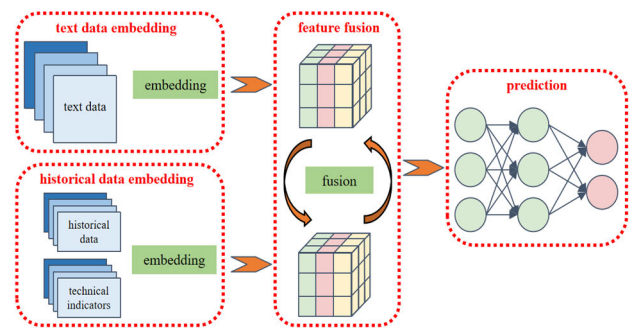


FIGURE 2. The flow of proposed method.

information loss will ensue. To address this limitation, we introduce Sequence Reorganization Text Embedding-BERT (SRTE-BERT), refer to Fig. 3, an approach designed to adopt extended textual sequences while preserving semantic integrity.

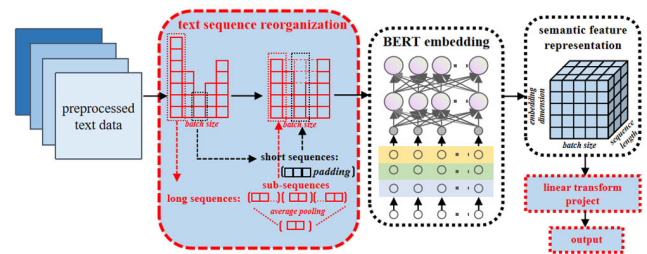


FIGURE 3. The proposed SRTE-BERT for text data embedding.

The text data Y is initially organized into batches based on the number of days d , resulting in varying sequence lengths across batches $[y_1, y_2, \dots, y_d]$, where y_α denotes the sequence corresponding to each day, with $\alpha \in \{1, d\}$. To address the issue of varying sequence lengths, the proposed method splits long sequences—those exceeding BERT's maximum sequence length of 512 tokens—into multiple sub-sequences. Assuming a long sequence y_{1s} , which is composed of sub-sequences $[ss_1, ss_2, \dots, ss_n]$, which ss_β denotes each sub-sequence in the long sequence y_{1s} , with

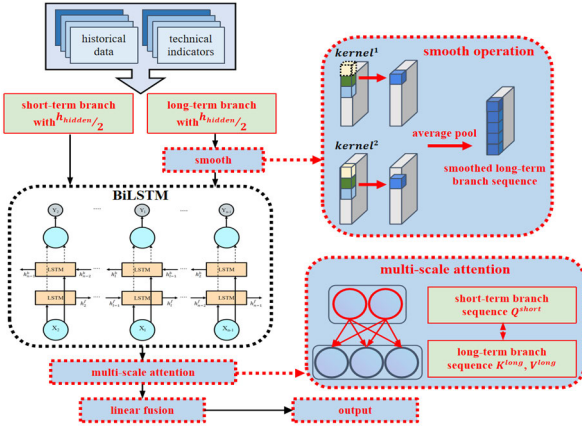


FIGURE 4. The proposed multi-scale LSTM for historical data embedding.

$\beta \in \{1, n\}$, and n representing the total number of sub-sequences. The number of sub-sequences n is calculated by dividing the length of y_{ls} by 512 and rounding up. If the last sub-sequence ss_n contains fewer than 512 tokens, it is padded to ensure that all sub-sequences within y_{ls} have equal lengths. Additionally, y_{ls} can be represented as a three-dimensional tensor, where γ_{ls}^0 denotes the original semantic representation of this long sequence (see Equation (1)).

$$y_{ls} = [ss_1, ss_2, \dots, ss_n] = (512, n, \gamma_{ls}^0) \quad (1)$$

Each sub-sequence ss_β in y_{ls} is represented as a two-dimensional tensor, where γ_{ss} denotes its semantic representation. The sub-sequence ss_β is determined using the following equation:

$$ss_\beta = (512, \gamma_{ss}), \quad \beta \in \{1, n\} \quad (2)$$

The sub-sequences are subsequently integrated into a single sequence using average pooling (see Equation (3)).

$$\begin{aligned} \text{AveragePool}[y_{ls}] &= \text{AveragePool}[ss_1, ss_2, \dots, ss_n] \\ &= (512, \gamma_{ls}^{ap}) \end{aligned} \quad (3)$$

where γ_{ls}^{ap} represents the semantic representation of the long sequence after applying average pooling along the dimension corresponding to the number of sub-sequences. For shorter sequences, automatic padding is applied to align their lengths with those of the processed long sequences, facilitating consistent representation across all sequences.

The BERT output, initially represented as a three-dimensional semantic feature $S_{BERT} = \text{BERT}_{\text{output}}(B_{bs}, B_{sl}, B_{ed})$, where B_{bs} denotes the batch size, B_{sl} represents the sequence length (fixed at 512), and B_{ed} indicates the embedding dimensions, is subsequently go through linear transform project to align the representational dimension with that of the historical price feature. The linear transformation here uses layer normalization and ReLU activation function (see Equation (4)), which enables this process to enhance the

expressive power while standardizing the feature distribution.

$$X_{proj}^{Bert} = \text{LayerNorm}(\text{ReLU})(W_{proj} \times X^{Bert} + b_{proj}) \quad (4)$$

where $W_{proj} \in \mathbb{R}^{d_{\text{hidden}} \times d_{Bert}}$ and $b_{proj} \in \mathbb{R}^{d_{\text{hidden}}}$. This transformation aligns the representational dimension of two different features in the embedding process.

2) MULTI-SCALE LONG SHORT-TERM MEMORY FOR HISTORICAL DATA EMBEDDING

The proposed architecture employs a multi-scale LSTM approach, refer to Fig. 4, to capture both short-term and long-term temporal dependencies in historical data. This architecture is designed to capture multiple time scales while capturing more complete contextual information, i.e., dealing with both short-term fluctuations and long-term trends in historical data. The details of the proposed MsLSTM are described as follows.

Given the input price sequence with positional encoding $X^{\text{price}} = [x_1, x_2, \dots, x_T]$, where T represents the sequence length, the multi-scale feature extraction is performed through parallel LSTM networks operating at different temporal scales. In this extraction process, the original sequence is divided into a short-term branch and a long-term branch. The short-term branch is used to process the complete original sequence and capture fine-grained short-term patterns (see Equation(5)). The long-term branch is used to process the time-smoothed sequence and focus on long-term trends.

$$\begin{aligned} \text{LSTM}_{\text{short}}(X^{\text{price}}) &= [h_1^{\text{short}}, h_2^{\text{short}}, \dots, h_T^{\text{short}}] \\ &\in \mathbb{R}^{Q \times h_{\text{hidden}}/2} \end{aligned} \quad (5)$$

where h_{hidden} denotes the hidden dimension in BiLSTM. For long-term branches, a fixed-size kernel k^s is used to calculate the moving average and form a new sequence with a stride of 1 (see Equation(6)). The 1 size stride is to ensure that the length of the long-term branch sequence is the same as the short-term branch sequence and the original sequence. The larger the kernel size, the better it can take into account long-term information, but it will lead to information redundancy. The length of the smoothed long-term branch sequence (see Equation (7)) is the same as the short-term branch and the original sequence length.

$$\begin{aligned} X^{\text{smooth}} &= \text{AvgPool}(X^{\text{price}}, k^s, \text{stride}), k^s \\ &< \text{sequence length} \end{aligned} \quad (6)$$

$$\begin{aligned} \text{LSTM}_{\text{long}}(X^{\text{smooth}}) &= [h_1^{\text{long}}, h_2^{\text{long}}, \dots, h_T^{\text{long}}] \\ &\in \mathbb{R}^{T \times h_{\text{hidden}}/2} \end{aligned} \quad (7)$$

To preserve temporal causality and prevent information leakage, the average pooling operation employs causal padding, ensuring that the smoothed value at each time step t only incorporates historical observations from $[t-k^s+1, t]$. This guarantees that no future information is used in computing the long-term branch features, which is essential for realistic forecasting performance evaluation.

The new long-term branch sequences and short-term branch sequences obtained are fused through the attention mechanism. (see Equation(8)). The multi-scale features are fused through a linear transformation (see Equation(9)). This design allows the model to adaptively select relevant information from long-term trends based on the needs of short-term patterns.

$$H^{attended} = \text{MultiHeadAttention}(Q^{short}, K^{long}, V^{long}) \quad (8)$$

$$H^{fused} = \text{LayerNorm}(W^f \times \text{Concat}(H^{short}, H^{attended})) + b_f \quad (9)$$

where W^f and b_f are learnable parameters. This concatenated fusion preserves the full information of short-term features while introducing attention-weighted long-term context.

3) CROSS-MODAL CROSS-FUSION FOR HETEROGENEOUS FEATURE FUSION

To address the challenge of effectively synthesizing heterogeneous feature representations, we introduce a CMCF method, which is structured into two complementary components and several sub-components, refer to Fig. 5, each optimized for particular fusion operations.

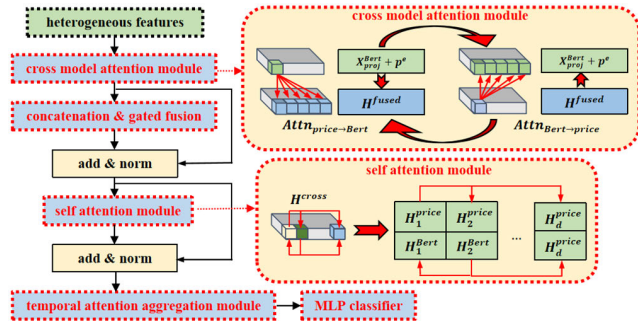


FIGURE 5. The proposed CMCF model for feature fusion.

The cross-modal fusion module integrates price sequence features with BERT-encoded textual features through a bidirectional attention mechanism. The BERT features $X^{Bert} \in \mathbb{R}^{Q \times d_{Bert}}$ are first projected with positional encoding, while the bidirectional cross-attention is computed using the following equation:

$$\text{Attn}_{price \rightarrow Bert}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (10)$$

where $Q = H^{fused}$, $K = V = X^{Bert}_{proj}$, and $d_k = d_{hidden}/h$, with h being the number of attention heads. Similarly, the reverse attention is computed using the following equation:

$$\text{Attn}_{Bert \rightarrow price}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (11)$$

where $Q = X^{Bert}_{proj}$, $K = V = H^{fused}$. The processed features from both attention directions are combined using a gated fusion mechanism using the following equations:

$$G = \sigma\left[W_g \times \text{Concat}\left(H^{price}, H^{Bert}\right) + b_g\right] \quad (12)$$

$$H^{cross} = G \odot H^{price} + (1 - G) \odot H^{Bert} \quad (13)$$

where σ denotes the sigmoid activation function, $\odot$ denotes element-wise multiplication, and $G \in \mathbb{R}^{Q \times d_{hidden}}$ serves as the gating weights. The fused features undergo self-attention to capture intra-sequence dependencies using the following equations:

$$(H^{LF}, A^{LF}) = \text{MultiHeadAttention}(H^{cross}, H^{cross}, H^{cross}) \quad (14)$$

$$H^{attended} = \text{LayerNorm}(H^{LF} + H^{cross}) \quad (15)$$

where H^{LF} denotes self-attention output, A^{LF} denotes self-attention weights, H^{cross} denotes the cross-modal fused features, and $H^{attended}$ denotes the self-attended processed features. A temporal attention mechanism is then applied to aggregate the sequence-level information (see Equation(16) and Equation(17)).

$$\alpha_t = \text{Softmax}\left[\tanh\left(W_\alpha \times h_t^{attended} + b_\alpha\right) \times v_\alpha\right] \quad (16)$$

$$H^{Final} = \sum_{t=1}^T \alpha_t \times h_t^{attended} \quad (17)$$

where H^{Final} denotes the final aggregated features, $W_\alpha \in \mathbb{R}^{d_{hidden}/2 \times d_{hidden}}$, $b_\alpha \in \mathbb{R}^{d_{hidden}/2}$, and $v_\alpha \in \mathbb{R}^{d_{hidden}/2}$. Compared to simple dimensionality reduction methods, this module can better retain important information and provide fixed-dimensional input for subsequent classifiers. The final prediction is obtained through a multi-layer perceptron with batch normalization and dropout regularization using the following equation:

$$\hat{y} = \sigma\left\{W_3 \times \text{ReLU}\left[W_2 \times \text{ReLU}\left(W_1 \times H^{Final} + b_1\right) + b_2\right] + b_3\right\} \quad (18)$$

where $W_1 \in \mathbb{R}^{d_{mlp} \times d_{hidden}}$, $W_2 \in \mathbb{R}^{d_{mlp}/2 \times d_{mlp}}$, and $W_3 \in \mathbb{R}^{d_{mlp}/4}$ are weight matrices with corresponding bias terms. The prediction classification is performed using a binary classifier. Specifically, the label is determined by comparing the predicted objective value v_i^e on the corresponding day i with the value v_{i-1}^e from the previous day as shown in Equation (19). The predicted value is obtained using Equation (20), and the loss is computed using the binary cross-entropy loss (BCELoss), as defined in Equation (21):

$$y = \begin{cases} 1, & v_i^e \geq v_{i-1}^e, \quad i \in \{1, d - \delta\} \\ 0, & v_i^e < v_{i-1}^e, \quad i \in \{1, d - \delta\} \end{cases} \quad (19)$$

$$\hat{y} = \sigma(FC_{output}^i), \quad i \in \{1, d - \delta\} \quad (20)$$

$$\text{BCE Loss}(\hat{y}, y) = -[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})] \quad (21)$$

where y represents the actual value, $\hat{y}$ is the predicted value, 1 indicates that the value increased compared to the previous day, and 0 indicates a decrease. Additionally, σ is the sigmoid function, and $\text{BCE Loss}(\hat{y}, y)$ represents the binary cross-entropy loss between the predicted value $\hat{y}$ and the actual value y .

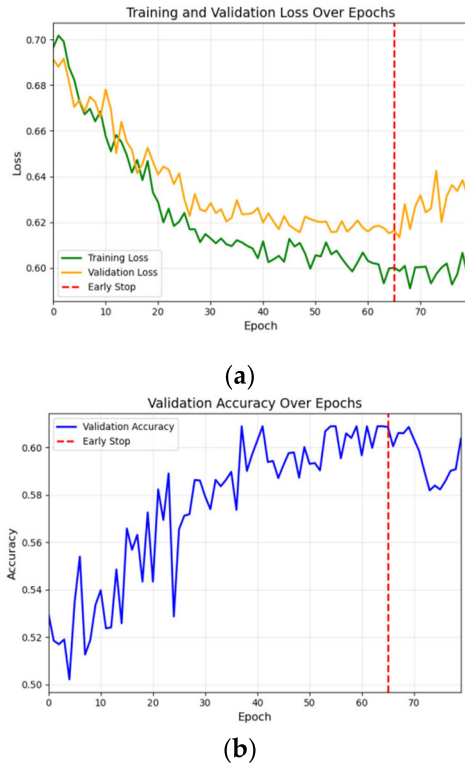


FIGURE 6. Training dynamics (a) and validation accuracy (b) under the optimal hyper-parameter configuration.

IV. RESULTS

A. EVALUATION METRICS

Accuracy (see Equation (22)), Precision (see Equation (23)), Recall (see Equation (24)), F1-score (see Equation (25)), and Matthews Correlation Coefficient (MCC) (see Equation (26)) are used as the primary evaluation metrics in this study.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (22)$$

$$Precision = \frac{TP}{TP + FP} \quad (23)$$

$$Recall = \frac{TP}{TP + FN} \quad (24)$$

$$F1Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (25)$$

MCC

$$= \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}} \quad (26)$$

Accuracy measures the proportion of correct predictions but can be misleading with imbalanced classes. The Matthews Correlation Coefficient (MCC) [30], [31] offers a more balanced alternative by incorporating all confusion matrix elements—TP, TN, FP, and FN. Precision measures the reliability of positive predictions, while recall measures coverage of actual positives. The F1-score combines both as their harmonic mean.

B. HYPERPARAMETER SETTING AND OPTIMIZATION

In this study, we adopt automated hyperparameter search methods. Specifically, Bayesian optimization utilizing Tree-structured Parzen Estimators (TPE) [32] is used in this study for hyperparameter tuning.

Epoch selection is often treated separately from the choice of hyperparameters, yet this separation can make large search spaces computationally expensive. Predefining a fixed number of epochs is also unreliable: the optimal training length can differ widely across model architectures, datasets, and hyperparameter settings. To overcome this problem, we use F1-score monitoring into the early stopping process. Specifically, F1-score on the validation set determines when training should stop in each hyperparameter optimization trial. The epoch count that produces the best configuration is then carried forward and used for evaluation on the test set.

Fig. 6 illustrates the training dynamics (a) and validation accuracy (b) under optimal hyperparameter configurations obtained through automated optimization. In terms of trends, the training and validation losses corresponding to the optimal hyperparameter configuration exhibited concurrent reduction prior to early stopping termination, with both values substantially below the theoretical minimum threshold of 0.693 for binary classification loss. Correspondingly, validation accuracy demonstrated progressive improvement until early stopping intervention.

TABLE 3. Hyperparameter, search range and value under automated optimization.

Hyperparameter	Search range	Value
Learning rate	1e-5 to 1e-2	0.002545
Dropout rate	0.1 to 0.5	0.310482
Batch size	32, 64, 128	32
Optimizer	Adam, AdamW, RMSprop, SGD	AdamW
Kernel size	3, 5, 7	7
Attention head	4, 8, 16	8
LSTM hidden dimension	64, 128, 256	64
MLP hidden dimension	64, 128, 256	128
Number of LSTM layers	1, 2, 3	2
Number of MLP layers	1, 2, 3	3
Early stop epoch	N/A	65

¹ N/A denotes not applicable.

Table 3 illustrates the best hyper-parameter configuration under automated optimization. Consequently, the comparative evaluation on the testing set will employ the optimal configuration.

C. BACK-TESTING ANALYSIS

To assess the practical applicability of our predictions, we conducted a back-testing analysis on the testing set. We implemented a simple long-only strategy: the model holds the asset when predicting an upward movement (label = 1) and stays out of the market otherwise (label = 0). Table 4 presents the key tradability metrics of our strategy compared

TABLE 4. Back-testing analysis results.

Metrics	Results	Baseline Results (Buy and Hold)
Cumulative Return	293.72% $\pm$ 85.70%	130.41%
Max Drawdown	-70.74% $\pm$ 8.19%	-87.25%
Turnover Rate	20.14% $\pm$ 6.19%	0%
Sharpe Ratio (annualized) ¹	0.30 $\pm$ 0.15	-
Win Rate (Accuracy)	58.09% $\pm$ 0.86%	-

¹Sharpe ratio calculated assuming risk-free rate = 0, following standard practice when rates are not specified for the evaluation period.

Note: Results averaged over 5 random seeds. Transaction costs and slippage are not included. Sensitivity analysis assuming 0.3% transaction cost suggests approximately 6% annualized cost drag.

to a buy-and-hold baseline. Our model achieved a maximum drawdown of $-70.74\% \pm 8.19\%$, which represents the worst-case peak-to-trough decline in cumulative returns. This is better than the buy-and-hold strategy's maximum drawdown of -87.25% , but still represents significant capital risk that may be unacceptable for many practical applications. The portfolio turnover rate was $20.14\% \pm 6.19\%$. This moderate level of turnover suggests that our model produces stable predictions rather than generating excessive trading signals. We also computed the annualized Sharpe ratio, which was 0.30 ± 0.15 . We assumed a risk-free rate of zero for this calculation, as: (1) risk-free rates vary significantly across the evaluation period and markets in our dataset, and (2) zero risk-free rate is the conventional baseline for strategy comparison when rates are not explicitly specified. The positive Sharpe ratio indicates that our strategy generates returns that compensate for the volatility risk undertaken. Our strategy achieved a cumulative return of $293.72\% \pm 85.70\%$ over the test period, outperforming the buy-and-hold baseline return of 130.41% . Besides, win rate (accuracy) was $58.09\% \pm 0.86\%$. These results suggest our model captures genuine predictive signals. However, the combination of substantial drawdown risk, modest Sharpe ratio, and potential real-world frictions (slippage, market impact, regime changes) indicate this should be viewed as a proof-of-concept demonstrating predictive capability rather than an immediately deployable trading system.

D. LONG-TEXT MODELING ANALYSIS

We conducted a comparative analysis encompassing token retention, computational efficiency, and predictive performance across our dataset of 42,840 daily text samples from 85 stocks. Our analysis reveals that 98.4% of samples contain ≤ 512 tokens, falling within BERT's native capacity. However, the conventional BERT truncation approach discards any content beyond this limit, resulting in a retention rate of only 82.18% (1,857,661 out of 2,260,364 tokens). In contrast, SRTE-BERT preserves 100% of textual content by segmenting longer sequences into multiple subsequences and applying average pooling, thereby retaining an additional 402,703 tokens (+17.82%) compared to direct truncation.

Importantly, only 0.4% of samples necessitate three or more subsequences, indicating that the additional computational overhead from sequence splitting affects a negligible portion of the dataset.

To address potential concerns regarding long-text modeling capabilities, we conducted controlled experiments comparing SRTE-BERT against two long-context encoders: Longformer [34] and BigBird [35]. Table 5 and 6 presents the comparison across predictive performance and computational metrics. All models were evaluated under identical conditions with frozen parameters to ensure fair comparison.

TABLE 5. Long-text embedding methods comparison results on prediction performance.

Model	Accuracy	F1-score	MCC
Longformer	0.5758 $\pm$ 0.0082	0.5709 $\pm$ 0.0141	0.1517 $\pm$ 0.0168
Bigbird	0.5812 $\pm$ 0.0089	0.5824 $\pm$ 0.0121	0.1624 $\pm$ 0.0178
SRTE-BERT	0.5809 $\pm$ 0.0086	0.5795 $\pm$ 0.0068	0.1623 $\pm$ 0.0167

TABLE 6. Long-text embedding methods comparison results on embedding efficiency.

Model	Embedding time (s)	Throughput (samples/s)	GPU memory (MB)
Longformer	5380.57	7.96	429.22
Bigbird	11188.51	3.83	2396.20
SRTE-BERT	532.92	80.39	29.14

The results demonstrate that while Longformer and BigBird achieve comparable accuracy, they incur drastically higher computational costs. Specifically, Longformer requires $10.1\times$ longer encoding time with $14.7\times$ greater GPU memory consumption. BigBird exhibits even more severe overhead at $21.0\times$ encoding time and $82.3\times$ memory usage. This translates to throughput reductions from 80.39 samples/second (SRTE-BERT) to merely 7.96 and 3.83 samples/second for Longformer and BigBird, respectively. For a dataset where 98.4% of samples fit within BERT's 512-token window, the $10\text{--}21\times$ computational penalty of long-context models yields negligible performance gains ($<0.5\%$ accuracy difference). The extreme memory requirements of BigBird render it impractical for production deployment and limit experimental iteration capacity. Furthermore, even the 1.6% of samples requiring truncation under standard BERT are adequately handled by SRTE-BERT's average pooling mechanism. The marginal performance differences across all three methods suggest that daily-aggregated social media sentiment exhibits high intra-day redundancy, where fine-grained long-range attention mechanisms provide minimal incremental value. This observation aligns with the domain characteristics of financial Twitter data: individual tweets are semantically independent units, and consensus sentiment is more predictive than nuanced sequential dependencies.

E. INTERPRETABILITY ANALYSIS

To provide interpretability insights, we conduct analysis of modality contributions and attention patterns. We quantify the contribution of each modality using a gate-based approach. For each sample in the test set, the fusion gate $g \in [0, 1]$ at time step t determines the weighting between price-enhanced and text-enhanced features. The modality contribution is computed as:

$$\text{Price Contribution} = \frac{1}{N} \sum_{i=1}^N \frac{1}{T \times d} \sum_{t=1}^T \sum_{j=1}^d g_i^{(t,j)} \quad (27)$$

$$\text{Text Contribution} = 1 - \text{Price Contribution} \quad (28)$$

where N denotes the number of samples, T denotes sequence length (time steps), d denotes the feature dimension, and $g_i^{(t,j)}$ denotes the gated sample for sample i at time t and dimension j . To ensure robustness, we calculate modality contributions across 5 independent runs with different random seeds. The results demonstrate that the price modality contributes 0.544 ± 0.063 while the text modality contributes 0.456 ± 0.063 to the final predictions. These results indicate a relatively balanced fusion strategy with slight preference for price information. The small standard deviation (0.063) demonstrates the stability of our fusion mechanism across different initializations and the near-equal contributions validate our design hypothesis that both modalities carry complementary information for the prediction.

Fig. 7 presents cross-modal attention patterns for two representative samples with positive predictions (label=1, $T^l=10$ and $T^l=30$). Several consistent patterns emerge: The attention matrices display clear lower-triangular structures, ensuring causal constraints where each time step attends only to historical information. Attention weights concentrate along diagonal and near-diagonal regions, revealing temporal proximity bias where the model prioritizes recent and corresponding cross-modal information. Both price-to-text and text-to-price directions exhibit symmetric attention patterns, indicating balanced bidirectional information flow. These patterns remain stable across different sequence lengths.

Fig. 8 presents temporal importance distributions for two positive prediction samples (label=1, $T^l=10$ and $T^l=30$). Both demonstrate strong recency bias: the most recent time steps receive the highest attention weights, while earlier time steps contribute progressively less. This distribution aligns with financial market dynamics where recent information holds greater predictive power for short-term movements. The more pronounced recency effect in shorter sequences suggests that relative temporal proximity is critical for effective cross-modal fusion.

F. ABLATION STUDY

To conduct a profound analysis of the effectiveness of our proposed method, we partition it into several sub-models to test their performance separately, each described as follows.

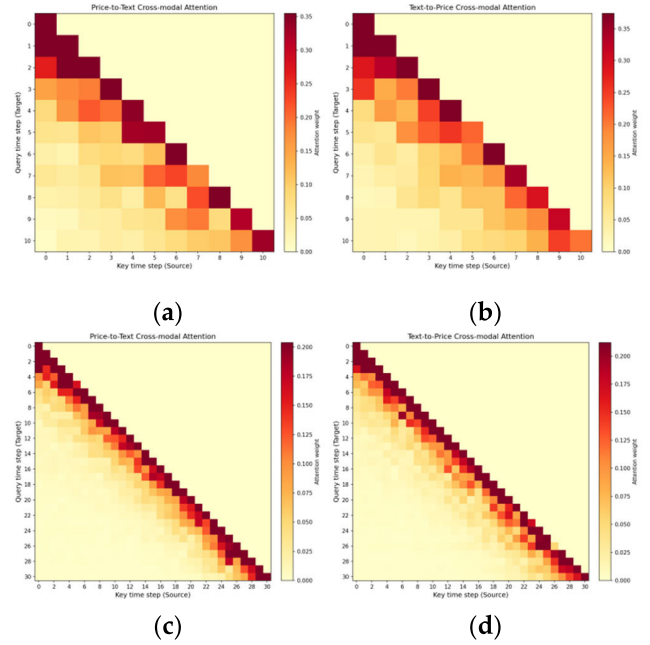


FIGURE 7. Cross-modal attention patterns for positive samples (label=1). (a-b) Sample with sequence length $T^l = 10$: (a) Price-to-text attention, (b) Text-to-Price attention. (c-d) Sample with sequence length $T^l = 30$: (c) Price-to-text attention, (d) Text-to-price attention.

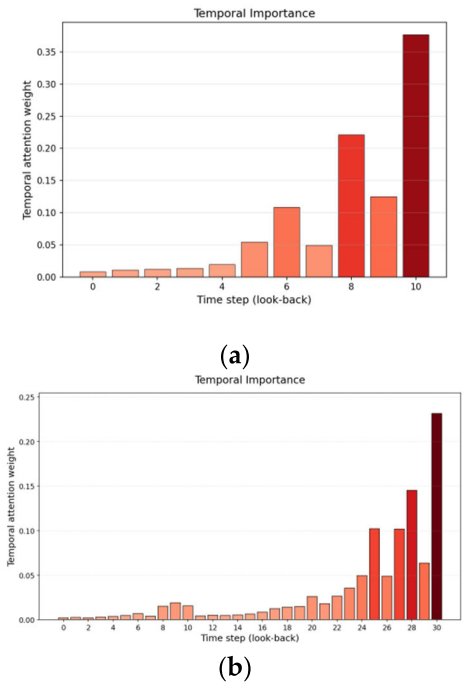


FIGURE 8. Temporal importance visualization for positive samples (label = 1) with different lag time ($T^l = 10$ and $T^l = 30$).

- LSTM: Using a simple LSTM architecture and the original five variables from the historical stock data, i.e., opening price (O), closing price (C), high price (H), low price (L), and trading volume (V).

TABLE 7. Performance comparison with sub-models in ablation experiment.

Category	Sub-model	Accuracy	Precision	Recall	F1	MCC
Single-modal	LSTM	0.5044 ± 0.0130	0.5080 ± 0.0146	0.4918 ± 0.0180	0.4958 ± 0.0113	0.0474 ± 0.0047
		0.5222 ± 0.0164	0.5259 ± 0.0143	0.5027 ± 0.0137	0.5037 ± 0.0127	0.0575 ± 0.0040
	LSTM+TI	0.5279 ± 0.0076	0.5412 ± 0.0083	0.5159 ± 0.0184	0.5241 ± 0.0121	0.0681 ± 0.0012
		0.5165 ± 0.0086	0.5225 ± 0.0129	0.5212 ± 0.0098	0.5218 ± 0.0080	0.0329 ± 0.0175
Multi-Modal	BERT	0.5177 ± 0.0177	0.5193 ± 0.0185	0.4927 ± 0.0145	0.4972 ± 0.0112	0.0782 ± 0.0032
		0.5719 ± 0.0094	0.5612 ± 0.0067	0.5819 ± 0.0176	0.5713 ± 0.0090	0.1444 ± 0.0192
	SRTE-BERT	0.5809 ± 0.0086	0.5826 ± 0.0166	0.5771 ± 0.0143	0.5795 ± 0.0068	0.1623 ± 0.0167
		0.5809 ± 0.0086	0.5826 ± 0.0166	0.5771 ± 0.0143	0.5795 ± 0.0068	0.1623 ± 0.0167

- LSTM +TI: Using a simple LSTM architecture and the original five variables. Meanwhile, technical indicators are added to the original variables of the historical stock data.
- MsLSTM +TI: Using proposed architecture to embed historical price data with technical indicators. The positional encoding part is removed from MsLSTM because no attention mechanism is used here for deep fusion.
- MsLSTM-BERT: The textual information is embedded using BERT's original embedding, and simple concatenation is used to fuse heterogeneous features.
- MsLSTM-(SRTE-BERT): The BERT inputs are embedded using our proposed method, and simple concatenation is used to fuse heterogeneous features.
- BERT-CMCF: Fusion of heterogeneous features is conducted using the proposed method instead of the simple concatenation method, but the text embedding section retains the embedded form of BERT.
- (SRTE-BERT)-CMCF: Integration of the proposed method.

We made two different control groups in the ablation study. The initial control group consisted of three sub-models, which used only historical and technical features as input. In contrast, the sub-models in the second control group incorporated heterogeneous data sources, specifically combining historical market data with textual information. The results of both experimental configurations are presented in TABLE 7.

The results show that the integrated method significantly outperforms other methods according to all the five evaluation criteria. This conclusion further verifies that every module plays a crucial role in effectively fusing heterogeneous data for stock movement prediction.

In the first control group (single-modal input), adding technical indicators and enhancing the predictive model's design (replacing conventional LSTM with multiscale LSTM) improved accuracy remarkably by 3.53% and 4.64%, respectively. Similar improvements were found in the other four performance measures. These findings are consistent with the recent direction of single-source historical data analysis

TABLE 8. Performance comparison with different lag time ((SRTE-BERT)-CMCF) in ablation experiment.

Lag time	Accuracy	Precision	Recall	F1	MCC
$T^l = 5$	0.5586 ± 0.0063	0.5647 ± 0.0241	0.5691 ± 0.0289	0.5666 ± 0.0233	0.1147 ± 0.0131
	0.5683 ± 0.0053	0.5738 ± 0.0108	0.5691 ± 0.0091	0.5714 ± 0.0066	0.1367 ± 0.0104
$T^l = 30$	0.5809 ± 0.0086	0.5826 ± 0.0166	0.5771 ± 0.0143	0.5795 ± 0.0068	0.1623 ± 0.0167

TABLE 9. Performance comparison with different lag time ((MsLSTM + TI) in ablation experiment.

Lag time	Accuracy	Precision	Recall	F1	MCC
$T^l = 5$	0.5239 ± 0.0062	0.5174 ± 0.0125	0.5243 ± 0.0049	0.5208 ± 0.0066	0.0476 ± 0.0123
	0.5367 ± 0.0106	0.5509 ± 0.0258	0.5378 ± 0.0185	0.5438 ± 0.0138	0.0740 ± 0.0206
$T^l = 30$	0.5279 ± 0.0076	0.5412 ± 0.0083	0.5159 ± 0.0184	0.5241 ± 0.0121	0.0681 ± 0.0012

TABLE 10. Performance comparison with baseline models.

Model	Accuracy	Precision	Recall	F1	MCC
Ensemble [11]	0.5246 ± 0.0042	0.5192 ± 0.0030	0.5458 ± 0.0103	0.5271 ± 0.0486	0.0512 ± 0.0121
Atten Bi-LSTM [10]	0.5384 ± 0.0017	0.5362 ± 0.0032	0.5377 ± 0.0304	0.5365 ± 0.0140	0.0769 ± 0.0033
TMFS	0.5541 ± 0.0031	0.5302 ± 0.0002	0.5586 ± 0.0014	0.5440 ± 0.0091	0.0885 ± 0.0063
LSTM+Fin	0.5291 ± 0.0059	0.5283 ± 0.0056	0.5093 ± 0.0746	0.5129 ± 0.0544	0.0593 ± 0.0126
BERT [6]	0.5128 ± 0.0036	0.5119 ± 0.0048	0.4629 ± 0.0461	0.4795 ± 0.0590	0.0256 ± 0.0074
HIMM [21]	0.5675 ± 0.0137	0.5644 ± 0.0296	0.5749 ± 0.0262	0.5693 ± 0.0240	0.1350 ± 0.0270
StockNet [22]	0.5809 ± 0.0086	0.5826 ± 0.0166	0.5771 ± 0.0143	0.5795 ± 0.0068	0.1623 ± 0.0167
Proposed	0.5809 ± 0.0086	0.5826 ± 0.0166	0.5771 ± 0.0143	0.5795 ± 0.0068	0.1623 ± 0.0167

research [33], which adopts two optimization directions: adding more input features and increasing model complexity, thus mining latent and unobvious temporal patterns.

The result in the second control group shows the fusion strategy choice strongly influences prediction performance. Specifically, the application of our proposed CMCF fusion technique achieves remarkable accuracy enhancements of $\approx 10.73\%$ and $\approx 12.21\%$ in the BERT- and SRTE-BERT-based sub-models, respectively, and these improvements are consistent across the four other evaluation metrics. The sub-models with the simple concatenation-based fusion (BERT and SRTE-BERT), however, show inferior predictive performance than the baseline model with only historical data and technical indicators (MsLSTM +TI), with the same setting as other conditions tested in our experiments. Results reveal that the insufficient integration, especially simple concatenation, injects a large amount of noises into the historical data, making it difficult to infer the weak temporal patterns,

and results in imbalanced dimensions of the different feature sets. Moreover, such architectures do not have the capability to make good use of semantic features derived from textual embeddings and are not able to access them to attach and interact with historical market data. Our proposed CMCF fusion resolves these problems and achieves significantly better predictive performance.

Experiments based on the comparative study of the text data embedding methods reveal that SRTE-BERT further improves the accuracy by $\approx 1.57\%$ compared to the normal BERT embeddings. This result further demonstrates the necessity to keep full semantic representations in deep fusion tasks.

We further tested the effect of different lag times T^l on the prediction performance of the proposed method. In the dataset construction, we did continuous lag modeling with equal intervals, so the range of T^l and input sequence length are equivalent. We selected three different sets of lag times with reference to previous studies, which are 5 [22], 10, and 30 [21].

Table 8 presents the different model performances of the proposed method at three different sets of lag times. The results show that the model achieves the most superior performance in the prediction task with long lag time ($T^l = 30$). To further extend this finding, we tested the MsLSTM + TI sub-model performance under different lag times (refer to Table 9). We did not include others because other sub-models (LSTM and LSTM + TI) have different variable inputs. The other sub-models using heterogeneous features (BERT, SRTE-BERT, and BERT-CMCF) on the fusion method presented a significant effect on the model performance.

According to the results, optimal model performance was achieved when $T^l = 10$. Subsequent extensions of the time lag beyond this threshold yielded negligible improvements in model efficacy. These results indicate that the incorporation of heterogeneous feature sets, coupled with attention-based fusion method—particularly inter-temporal attention mechanisms—enables the model to selectively focus on temporally critical observations.

G. BASELINE METHODS AND COMPARISON RESULT

By comparing our proposed method's predictive capabilities with other methods, we conduct a comparative experiment using the same StockNet dataset. The details of the baseline methods are described as follows.

- **Ensemble [11]:** An ensemble method uses only historical price data to effectively integrate linear and nonlinear features for stock market prediction.
- **Atten Bi-LSTM [10]:** A Bi-directional LSTM structure combined with an attention mechanism is used to obtain richer contextual information in historical stock data.
- **TMFS Multifactor [16]:** An early fusion model using TextBlob to extract sentiment scores from news data and combined with raw variables from stock data to accomplish predictions in an LSTM architecture.

- **LSTM + FinBERT [6]:** An early fusion model using pre-trained FinBERT through financial text to extract sentiment scores from news data combined with stock data raw variables to complete the prediction in an LSTM architecture.
- **Hybrid Information Mixing Module [21]:** A deep fusion model that retains the complete semantic representation of text data. The embedding of semantic and time series features is done by BERT and GRU, respectively, and the heterogeneous features are fused and predicted after completing the embedding.
- **StockNet [22]:** A deep generative model that introduces recurrent continuous latent variables through variational auto-encoders to capture market stochasticity, using neural variational inference to handle intractable posterior inference.

Table 10 illustrates the comparison of the predicted performance of the baseline methods and the proposed one, using the same dataset. The results indicate the superiority of the proposed method in making predictions in terms of various evaluation metrics. The results also validate that the predicted stock price trends are well aligned with the observed market trends and support that the proposed method achieves promising results over existing methods in the comparison setup.

V. CONCLUSION

In this paper, we proposed a multi-modal deep fusion method for stock movement prediction that jointly learns from the text and historical market data by three complementary modules: SRTE-BERT for semantic preservation, MsLSTM for multi-scale temporal modeling, and CMCF for dual-level heterogeneous feature fusion. Extensive experiments on the StockNet dataset indicated that our model achieves $\approx 58.09\%$ accuracy, $\approx 57.95\%$ F1-score, and ≈ 0.1623 MCC, which are significantly better than selected baseline methods. Additionally, these findings confirm the viability of deep multimodal fusion for modeling intricate financial relations and the potential to substantially improve the reliability of financial forecasting in real-world markets. The ablation analysis validated the importance of each component, especially the advantage of CMCF over simple concatenation and the necessity of SRTE-BERT in preserving the semantic richness. Furthermore, the investigation of lag configurations illustrated that medium- to long-range dependencies are necessary for achieving good prediction results. In the future, we will extend the framework to include various text embedding techniques and seek advanced fusion mechanisms on how to improve the generalization in dynamic and volatile financial scenarios.

Our long-text modeling analysis demonstrates the effectiveness of the SRTE-BERT approach in preserving semantic completeness while maintaining computational efficiency. The analysis on 42,840 daily samples reveals that SRTE-BERT retains 100% of textual content (2,260,364 tokens) compared to only 82.18% retention with conventional

BERT truncation, recovering an additional 402,703 tokens. Controlled experiments against specialized long-context encoders confirm that SRTE-BERT achieves comparable performance while requiring significantly less computational overhead. This validates our design hypothesis that average pooling over segmented sequences effectively preserves semantic information for financial prediction tasks without the computational burden of full long-context attention mechanisms. Despite its efficacy in preserving textual integrity, the current SRTE-BERT module relies on an average pooling mechanism to aggregate sub-sequence embeddings. A potential limitation of this approach is signal dilution: in scenarios where a long document contains a single, critical signal (e.g., a “market crash” warning) in significant neutral noise, simple averaging may “wash out” the intensity of such sparse but vital information. While our current design prioritizes computational throughput, future work will investigate hierarchical attention mechanisms to adaptively weight sub-sequences based on their semantic importance, thereby ensuring that localized critical signals are effectively preserved within the global representation. Furthermore, the MsLSTM architecture currently extracts long-term market trends using a 1D average pooling operation. Although this provides a stable linear low-pass filter for capturing general trends while maintaining sequence alignment, it may be considered rudimentary for modeling highly complex, non-linear long-term dependencies in financial time series. The use of simple moving averages can inherently introduce phase lag, which might slightly delay the model’s response to rapid regime shifts. To address this, future iterations of the framework could incorporate more advanced non-linear modeling techniques, such as Dilated Convolutions or Temporal Convolutional Networks, to capture intricate multi-scale patterns without the constraints of linear smoothing.

The interpretability analysis reveals balanced and meaningful cross-modal fusion patterns in our framework. Quantitative modality contribution analysis across 5 independent runs shows that price and text modalities contribute ≈ 0.544 and ≈ 0.456 respectively, indicating a relatively balanced fusion strategy that demonstrates stability across initializations. Attention pattern visualization uncovers several key characteristics: (1) clear lower-triangular structures ensuring proper causal constraints, (2) concentration along diagonal regions indicating temporal proximity bias where recent cross-modal information receives higher weight, (3) symmetric bidirectional attention patterns reflecting balanced information flow between modalities, and (4) strong recency bias in temporal importance distributions that aligns with financial market dynamics. These patterns validate that our CMCF mechanism effectively captures complementary information from both modalities while maintaining interpretable and financially-grounded attention distributions.

Back-testing results suggest our model captures genuine predictive signals. However, the combination of substantial drawdown risk, modest Sharpe ratio, and potential real-world frictions (slippage, market impact, regime changes) indicate

this should be viewed as a proof-of-concept demonstrating predictive capability rather than an immediately deployable trading system. We acknowledge that movement prediction does not capture the magnitude of price changes. Future extensions could incorporate multi-task learning to simultaneously predict both direction and magnitude, or adopt ordinal classification to capture finer-grained patterns while maintaining the robustness advantages of classification.

ABBREVIATIONS

The following abbreviations are used in this manuscript:

BERT	Bi-directional Encoder Representation from Transformer.
Bi-LSTM	Bi-directional Long Short-Term Memory.
C	Closing Price.
CMCF	Cross-Modal Cross-Fusion.
GRU	Gated Recurrent Unit.
H	High Price.
L	Low Price.
LSTM	Long Short-Term Memory.
MA	Moving Average.
MACD	Moving Average Convergence/Divergence.
MCC	Matthews Correlation Coefficient.
MOM	Momentum.
MsLSTM	Multi-scale Long Short-Term Memory.
O	Opening Price.
OBV	On-balance Volume.
RNN	Recurrent Neural Network.
ROC	Rate of Change.
RSI	Relative Strength Index.
SRTE	Sequence Reorganization Text Embedding.
TI	Technical Indicators.
TPE	Tree-structured Parzen Estimators.
V	Trading Volume.
VPT	Volume Price Trend.

DATA AVAILABILITY

The original dataset can be found at: <https://github.com/yumoxu/stocknet-dataset>. Our model code and preprocessed data can be found at: <https://github.com/Veecky29/MsLSTM-SRTE-BERT-CMCF>.

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REFERENCES

- [1] J. Shah, D. Vaidya, and M. Shah, “A comprehensive review on multiple hybrid deep learning approaches for stock prediction,” *Intell. Syst. with Appl.*, vol. 16, Nov. 2022, Art. no. 200111, doi: 10.1016/j.iswa.2022.200111.
- [2] W. Khan, M. A. Ghazanfar, M. A. Azam, A. Karami, K. H. Alyoubi, and A. S. Alfakeeh, “Stock market prediction using machine learning classifiers and social media, news,” *J. Ambient Intell. Humanized Comput.*, vol. 13, no. 7, pp. 3433–3456, Jul. 2022, doi: 10.1007/s12652-020-01839-w.

- [3] B. A. Abdelfattah, S. M. Darwish, and S. M. Elkaffas, "Enhancing the prediction of stock market movement using neutrosophic-logic-based sentiment analysis," *J. Theor. Appl. Electron. Commerce Res.*, vol. 19, no. 1, pp. 116–134, Jan. 2024, doi: [10.3390/jtaer19010007](#).
- [4] A. Gumus and C. O. Sakar, "Stock market prediction by combining stock price information and sentiment analysis duygulari analizi ve hisse fiyat bilgisinin birleştirilmesiyle hisse senedi piyasası tahmini," *Int. J. Adv. Eng. Pure Sci.*, vol. 33, no. 1, pp. 18–27, Jan. 2021, doi: [10.7240/jeps.683952](#).
- [5] T. Julian, T. Devrison, V. Anora, and K. M. Suryaningrum, "Stock price prediction model using deep learning optimization based on technical analysis indicators," *Procedia Comput. Sci.*, vol. 227, pp. 939–947, Jan. 2023, doi: [10.1016/j.procs.2023.10.601](#).
- [6] J. Kim, H.-S. Kim, and S.-Y. Choi, "Forecasting the S&P 500 index using mathematical-based sentiment analysis and deep learning models: A FinBERT transformer model and LSTM," *Axioms*, vol. 12, no. 9, p. 835, Aug. 2023, doi: [10.3390/axioms12090835](#).
- [7] C. Zhang, N. N. A. Sjarif, and R. B. Ibrahim, "Decision fusion for stock market prediction: A systematic review," *IEEE Access*, vol. 10, pp. 81364–81379, 2022, doi: [10.1109/ACCESS.2022.3195942](#).
- [8] L. N. Mintarya, J. N. M. Halim, C. Angie, S. Achmad, and A. Kurniawan, "Machine learning approaches in stock market prediction: A systematic literature review," *Procedia Comput. Sci.*, vol. 216, pp. 96–102, Jan. 2023, doi: [10.1016/j.procs.2022.12.115](#).
- [9] C. Y. Lin and J. A. L. Marques, "Stock market prediction using artificial intelligence: A systematic review of systematic reviews," *Social Sci. Humanities Open*, vol. 9, Jun. 2024, Art. no. 100864, doi: [10.1016/j.ssaho.2024.100864](#).
- [10] W. Liu, Y. Zhang, and Y. Liu, "Attention-based BiLSTM model for stock price prediction," in *Proc. 5th Int. Conf. Artif. Intell. Pattern Recognit.*, Xiamen, China, Sep. 2022, pp. 257–263.
- [11] K. He, Q. Yang, L. Ji, J. Pan, and Y. Zou, "Financial time series forecasting with the deep learning ensemble model," *Mathematics*, vol. 11, no. 4, p. 1054, Feb. 2023, doi: [10.3390/math11041054](#).
- [12] R. Ho and K. Hung, "CEEMD-based multivariate financial time series forecasting using a temporal fusion transformer," in *Proc. IEEE 14th Symp. Comput. Appl. Ind. Electron. (ISCAIE)*, Penang, Malaysia, May 2024, pp. 209–215.
- [13] A. Lazcano, P. J. Herrera, and M. Monge, "A combined model based on recurrent neural networks and graph convolutional networks for financial time series forecasting," *Mathematics*, vol. 11, no. 1, p. 224, Jan. 2023, doi: [10.3390/math11010224](#).
- [14] N. Sathish and B. Jamalpur, "A comprehensive survey on enhancing stock market predictions through machine learning and NLP-based sentiment analysis integration," in *Proc. 4th Int. Conf. Sentiment Anal. Deep Learn. (ICSADL)*, Bhimadatta, Nepal, Feb. 2025, pp. 92–99.
- [15] K. Du, F. Xing, R. Mao, and E. Cambria, "Financial sentiment analysis: Techniques and applications," *ACM Comput. Surveys*, vol. 56, no. 9, pp. 1–42, Oct. 2024, doi: [10.1145/3649451](#).
- [16] A. Deep, "A multifactor analysis model for stock market prediction," *Int. J. Comput. Sci. Telecommun.*, vol. 4, no. 1, pp. 1–6, 2023.
- [17] A. John and T. Latha, "Stock market prediction based on deep hybrid RNN model and sentiment analysis," *Automatika*, vol. 64, no. 4, pp. 981–995, Oct. 2023, doi: [10.1080/00051144.2023.2217602](#).
- [18] W.-J. Liu, Y.-B. Ge, and Y.-C. Gu, "News-driven stock market index prediction based on trellis network and sentiment attention mechanism," *Expert Syst. Appl.*, vol. 250, Sep. 2024, Art. no. 123966, doi: [10.1016/j.eswa.2024.123966](#).
- [19] K. Fu and Y. Zhang, "Incorporating multi-source market sentiment and price data for stock price prediction," *Mathematics*, vol. 12, no. 10, p. 1572, May 2024, doi: [10.3390/math12101572](#).
- [20] F. Ronaghi, M. Salimibeni, F. Naderkhani, and A. Mohammadi, "ND-SMPF: A noisy deep neural network fusion framework for stock price movement prediction," in *Proc. IEEE 23rd Int. Conf. Inf. Fusion (FUSION)*, Jul. 2020, pp. 1–7.
- [21] J. Choi, S. Yoo, X. Zhou, and Y. Kim, "Hybrid information mixing module for stock movement prediction," *IEEE Access*, vol. 11, pp. 28781–28790, 2023, doi: [10.1109/ACCESS.2023.3258695](#).
- [22] Y. Xu and S. B. Cohen, "Stock movement prediction from tweets and historical prices," in *Proc. 56th Annu. Meeting Assoc. Comput. Linguistics (Volume 1: Long Papers)*, Jul. 2018, pp. 1970–1979.
- [23] M. R. Vargas, C. E. M. dos Anjos, G. L. G. Bichara, and A. G. Evsukoff, "Deep learning for stock market prediction using technical indicators and financial news articles," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Rio de Janeiro, Brazil, Jul. 2018, pp. 1–8.
- [24] M. Agrawal, P. Kumar Shukla, R. Nair, A. Nayyar, and M. Masud, "Stock prediction based on technical indicators using deep learning model," *Comput., Mater. Continua*, vol. 70, no. 1, pp. 287–304, 2022, doi: [10.32604/cmc.2022.014637](#).
- [25] S. K. Chandar, "Convolutional neural network for stock trading using technical indicators," *Automated Softw. Eng.*, vol. 29, no. 1, May 2022, Art. no. 16, doi: [10.1007/s10515-021-00303-z](#).
- [26] S. Albahli, T. Nazir, M. Nawaz, and A. Irtaza, "An improved DenseNet model for prediction of stock market using stock technical indicators," *Expert Syst. Appl.*, vol. 232, Dec. 2023, Art. no. 120903, doi: [10.1016/j.eswa.2023.120903](#).
- [27] A. T. Haryono, R. Sarno, and K. R. Sungkono, "Transformer-gated recurrent unit method for predicting stock price based on news sentiments and technical indicators," *IEEE Access*, vol. 11, pp. 77132–77146, 2023, doi: [10.1109/ACCESS.2023.3298445](#).
- [28] H. Lee, J. H. Kim, and H. S. Jung, "Deep-learning-based stock market prediction incorporating ESG sentiment and technical indicators," *Sci. Rep.*, vol. 14, no. 1, May 2024, Art. no. 10262, doi: [10.1038/s41598-024-61106-2](#).
- [29] M. V. Koroteev, "BERT: A review of applications in natural language processing and understanding," 2021, *arXiv:2103.11943*.
- [30] B. Xie, R. J. Passonneau, L. Wu, and G. G. Creamer, "Semantic frames to predict stock price movement," in *Proc. 51st Annu. Meeting Assoc. Comput. Linguistics (ACL)*, Sofia, Bulgaria, Aug. 2013, pp. 873–883.
- [31] X. Ding, Y. Zhang, T. Liu, and J. Duan, "Deep learning for event-driven stock prediction," in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, Buenos Aires, Argentina, Jul. 2015, pp. 2327–2333.
- [32] S. Tao, P. Peng, Y. Li, H. Sun, Q. Li, and H. Wang, "Supervised contrastive representation learning with tree-structured Parzen estimator Bayesian optimization for imbalanced tabular data," *Expert Syst. Appl.*, vol. 237, Mar. 2024, Art. no. 121294, doi: [10.1016/j.eswa.2023.121294](#).
- [33] T. Burzykowski, M. Geubbelmans, A.-J. Rousseau, and D. Valkenburg, "Validation of machine learning algorithms," *Amer. J. Orthodontics Dentofacial Orthopedics*, vol. 164, no. 2, pp. 295–297, Aug. 2023, doi: [10.1016/j.ajodo.2023.05.007](#).
- [34] I. Beltagy, M. E. Peters, and A. Cohan, "Longformer: The long-document transformer," 2020, *arXiv:2004.05150*.
- [35] M. Zaheer, G. Guruganesh, A. Dubey, J. Ainslie, C. Alberti, S. Ontanon, P. Pham, A. Ravula, Q. Wang, L. Yang, and A. Ahmed, "Big bird: Transformers for longer sequences," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Vancouver, BC, Canada, 2020, pp. 17283–17297.



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